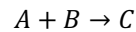


Practice: One Model, Different Perspectives

Ungraded practice • No submission required

The goal is to connect simple mathematical models with everyday and physiological scenarios, explore the same problem from different perspectives, develop critical-thinking skills, and learn to apply scientific thinking to a wide range of real-world and daily-life situations.

Consider the simple reaction:

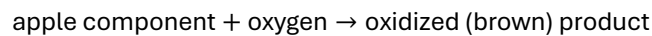


Using the **law of mass action**, the formation of C can be described as:

$$\frac{dC}{dt} = k[A][B]$$

We talked about this in class:

Imagine cutting an apple in half. After being exposed to air, the white surface gradually turns brown.

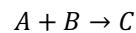


1. Now Think About Physiology

Imagine that a **cell or tissue is cut or injured**. What happens next?

There is no single answer. Think about the situation from your own perspective. You might consider **blood clotting, tissue repair, inflammation, cell signaling, immune responses**, or another process that interests you.

Try to describe **one part of the response** using:

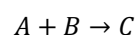


- What do A , B , and C represent in your interpretation?
- What does k represent?
- According to $\frac{dC}{dt} = k[A][B]$, what happens if $[A]$, $[B]$, or k increases?
- What aspects of the real physiological process does this simple model leave out?

2. Implement via AI-Assisted Coding

Choose the physiological interpretation you developed in **Part 1** (e.g., tissue repair).

Use an **AI-assisted coding tool** to implement the simple model



and simulate how $[A]$, $[B]$, and $[C]$ change over time.

Try changing the initial values of A and B , or the rate constant k .

Observe: How do these changes affect the behavior of your model? Does the result make sense for the physiological scenario you chose?